

Janitha Aswedige

janitha@hawaii.edu
LinkedIn
Personal Website

RESEARCH INTERESTS

- Formal methods and interactive theorem proving (Lean, Coq), with interests in proof theory, mechanized mathematics, and program verification.

SKILLS

- **Programming:** Lean 4, Mathlib 4, Coq, and Python (working knowledge).
- **Tools:** L^AT_EX, Git, and VS Code.
- **Areas:** Formal verification, theorem proving, and mathematical logic.

EDUCATION

University of Hawai'i at Mānoa, Honolulu, Hawai'i, United States. 2024
Master of Arts, Mathematics
Thesis Title: A Monadic Logic for Capacity Quantifiers
Advisor: Prof. David Ross

University of Colombo, Colombo, Sri Lanka. 2017
Bachelor of Science (Hons) Mathematics

ACADEMIC EXPERIENCE

Vals AI 2025
Research Collaborator

- Contributed to ProofBench, a benchmark evaluating LLMs on formally verified mathematics.
- Formalized graduate-level logic problems in Lean 4 for automated theorem proving evaluation.

Project Numina 2025
Contributor

- Contributed to Project Numina through Kili Technology, formalizing AIME and IMO problems in Lean 4 for mathematical reasoning datasets.

University of Hawai'i at Mānoa 2019-2023
Graduate Assistant

- Instructor of record for Calculus I (Summer).
- Led recitations for Calculus I & II (100+ students).
- Designed and graded worksheets and assessments.

University of Colombo 2018-2019
Assistant Lecturer

- Conducted tutorial classes for Calculus I & II, Differential Equations, Graph Theory, Analysis and Linear Algebra.
- Worked as the Coordinator for Financial Mathematics courses.
- Worked as the Assistant Coordinator for Pure Mathematics.

Publications

- **Janitha Aswedige**. A Monadic Logic for Capacity Quantifiers. *Mathematical Logic Quarterly*.
- Nikil Ravi, Kexing Ying, Vasilii Nesterov, Rayan Krishnan, Elif Uskuplu, Bingyu Xia, **Janitha Aswedige**, Langston Nashold. FormalProofBench: Can Models Write Graduate Level Math Proofs That Are Formally Verified? *VerifAI-2 Workshop, ICLR 2026*.
- **Janitha Aswedige**, Rajeev Goré. Corrigendum: A Correct Polynomial Translation Of S4 Into Intuitionistic Logic. In preparation.

TALKS

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| • <i>Filtrations and Finite Model Property of $S5$</i> | University of Hawai'i Logic Seminar, Fall 2021 |
| • <i>Inference Rules for Probability Logic by Marija Boricic</i> | University of Hawai'i Logic Seminar, Fall 2021 |
| • <i>A Monadic Logic for Capacity Quantifiers</i> | University of Hawai'i, Master's Thesis Defense |
| • <i>Capacity Quantifiers</i> | Oslo Logic Seminar 2026 |

SELECTED PROJECTS

University of Hawai'i

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| • <i>Modal Invariance - Class Project MATH 654 (Introduction to Logic)</i> | 2020 |
| I gave a talk on Modal Invariance based on Johan van Benthem's "Modal Logic for Open Minds." | |
| • <i>Formalizing some results related to ordered and unordered pairs in Lean - Class Project MATH 654</i> | 2020 |
| This was done as a group project for the final examination. | |
| • <i>Alpha Theory - Class project MATH 649B (Nonstandard Analysis)</i> | 2021 |
| I discussed the paper "Alpha-theory: An elementary axiomatics for nonstandard analysis" by Vieri Benci and Mauro Di Nasso as the final presentation for the Nonstandard Analysis course. Here, I compared the super-structure approach with Benci and Di Nasso's approach. | |
| • <i>Relevance and Detachment - MATH 699 (Modal Logic)</i> | 2021 |
| This was done as a reading course in deontic logic with Prof. Bjørn Kjos-Hanssen and R.J. Reiff. | |
| • <i>Arithmetical Hierarchy - Class Project MATH 649B (Computability Theory)</i> | 2022 |
| I gave a talk on the arithmetical hierarchy as a component of the final exam. | |

AWARDS AND HONOURS

Achievement Scholarship of \$800 for 2019-2020	University of Hawai'i at Mānoa
Achievement Scholarship of \$800 for 2020-2021	University of Hawai'i at Mānoa
Achievement Scholarship of \$250 for 2021-2022	University of Hawai'i at Mānoa
Research Scholarship - Math Department \$4000 2023	University of Hawai'i at Mānoa
Student Affiliate	East-West Center, Honolulu, Hawai'i

OTHER EXPERIENCES

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| • Organized an orientation session in math for incoming students | University of Colombo, Sri Lanka |
| • Organized a problem-solving session for elementary school students | University of Colombo, Sri Lanka |
| • Conducted Training Sessions for Sri Lanka Mathematics Competition | Sri Lanka Olympiad Mathematics Foundation |
| • Volunteered at the Community Building Institute | East-West Center, Honolulu, Hawai'i |
| • Volunteered at The Exchange | East-West Center, Honolulu, Hawai'i |

REFERENCES

Professor David Ross

Professor, Department of Mathematics, University of Hawai'i at Mānoa, HI, USA
E-mail: ross@math.hawaii.edu

Professor Bjørn Kjos-Hanssen

Professor, Department of Mathematics, University of Hawai'i at Mānoa, HI, USA
E-mail: bjoernkh@hawaii.edu